

Chapter 5. Text Clustering and Topic Modeling

```
# install these things
!pip install -U \
transformers==4.46.3 \
huggingface_hub==0.26.2 \
accelerate==1.1.1 \
tokenizers==0.20.3 \
safetensors
```

Requirement already satisfied: sympy<1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

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Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

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Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3)

Requirement already satisfied: cuda-pathfinder~1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12.9.4->torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.10.0->accelerate==1.1.1)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch>=1.10.0->accelerate==1.1.1)

Downloading transformers-4.46.3-py3-none-any.whl (10.0 MB)
10.0/10.0 MB 83.3 MB/s eta 0:00:00

Downloading huggingface_hub-0.26.2-py3-none-any.whl (447 kB)
447.5/447.5 kB 38.5 MB/s eta 0:00:00

Downloading accelerate-1.1.1-py3-none-any.whl (333 kB)
333.2/333.2 kB 21.9 MB/s eta 0:00:00

Downloading tokenizers-0.20.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.0 MB)
3.0/3.0 MB 89.4 MB/s eta 0:00:00

Installing collected packages: huggingface_hub, tokenizers, transformers, accelerate

Attempting uninstall: huggingface_hub
Found existing installation: huggingface_hub 1.23.0
Uninstalling huggingface_hub-1.23.0:
Successfully uninstalled huggingface_hub-1.23.0

Attempting uninstall: tokenizers
Found existing installation: tokenizers 0.22.2
Uninstalling tokenizers-0.22.2:
Successfully uninstalled tokenizers-0.22.2

Attempting uninstall: transformers
Found existing installation: transformers 5.13.1
Uninstalling transformers-5.13.1:
Successfully uninstalled transformers-5.13.1

Attempting uninstall: accelerate
Found existing installation: accelerate 1.14.0
Uninstalling accelerate-1.14.0:
Successfully uninstalled accelerate-1.14.0

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is deprecated. Use pip install --ignore-installed to force pip to ignore installed packages.
diffusers 0.39.0 requires huggingface-hub<2.0,>=0.34.0, but you have huggingface-hub 0.26.2 which is incompatible.
gradio 6.20.0 requires huggingface-hub<2.0,>=1.2.0, but you have huggingface-hub 0.26.2 which is incompatible.
Successfully installed accelerate-1.1.1 huggingface_hub-0.26.2 tokenizers-0.20.3 transformers-4.46.3

```
# Load data from Hugging Face
from datasets import load_dataset

# Download and load the "arxiv_nlp" training dataset
dataset = load_dataset("maartengr/arxiv_nlp")["train"]

# Get all the abstracts and titles from the dataset
abstracts = dataset["Abstracts"]
titles = dataset["Titles"]
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
README.md: 100% 617/617 [00:00<00:00, 23.5kB/s]
data.csv: 100% 53.2M/53.2M [00:10<00:00, 5.44MB/s]
Generating train split: 44949/0 [00:01<00:00, 38444.45 examples/s]

```

```

# A Common Pipeline for Text Clustering
# Embedding Documents

```

```

# Import the SentenceTransformer class
from sentence_transformers import SentenceTransformer

# Load the pre-trained GTE-Small embedding model
embedding_model = SentenceTransformer("thenlper/gte-small")

# Convert all abstracts into numerical vector embeddings
embeddings = embedding_model.encode(abstracts,
show_progress_bar=True # Show the embedding generation progress
)

print(embeddings.shape)

```

```

The cache for model files in Transformers v4.22.0 has been updated. Migrating your old cache. This is a one-time only operat
0/0 [00:00<?, ?it/s]

modules.json: 100% 385/385 [00:00<00:00, 10.1kB/s]
README.md: 68.1k/? [00:00<00:00, 3.66MB/s]
sentence_bert_config.json: 100% 57.0/57.0 [00:00<00:00, 3.00kB/s]
config.json: 100% 583/583 [00:00<00:00, 22.1kB/s]
model.safetensors: 100% 66.7M/66.7M [00:01<00:00, 46.7MB/s]
tokenizer_config.json: 100% 394/394 [00:00<00:00, 26.1kB/s]
vocab.txt: 232k/? [00:00<00:00, 6.34MB/s]
tokenizer.json: 712k/? [00:00<00:00, 36.8MB/s]
special_tokens_map.json: 100% 125/125 [00:00<00:00, 8.32kB/s]
config.json: 100% 190/190 [00:00<00:00, 17.7kB/s]
Batches: 100% 1405/1405 [03:50<00:00, 22.62it/s]
(44949, 384)

```

```

# Reducing the Dimensionality of Embeddings

```

```

# Import the UMAP class for dimensionality reduction
from umap import UMAP

# We reduce the input embeddings from 384 dimensions to 5 dimensions
umap_model = UMAP(
    n_components=5, # Reduce the embeddings to 5 dimensions
    min_dist=0.0, # Keep similar data points close together
    metric='cosine', # Use cosine distance to measure similarity
    random_state=42 # Set a fixed random seed for reproducible results
)

# Reduce the dimensions of the embeddings
reduced_embeddings = umap_model.fit_transform(embeddings)
print(reduced_embeddings)

```

```

/usr/local/lib/python3.12/dist-packages/umap/umap_.py:1952: UserWarning: n_jobs value 1 overridden to 1 by setting random_st
warn(
[[ 0.06644218  0.10430977  9.84145    0.9330931  3.3611138 ]
 [ 0.07065292  0.11036061  9.835484    0.9511084  3.3666885 ]
 [ 0.4720674  -0.0840879   9.358524    1.3132552  3.110296 ]
 ...
 [ 0.67552364 -0.3260398   9.123457    3.2283785  1.012194 ]
 [ 0.18272185 -0.60812855  9.591652    2.5482793  2.6015942 ]
 [ 0.68380034 -0.33928108  9.114251    3.2189822  1.0150725 ]]

```

```
# Cluster the Reduced Embeddings
```

```
# Import the HDBSCAN class for clustering
from hdbscan import HDBSCAN

# Create and train the HDBSCAN model to group similar embeddings into clusters
hdbscan_model = HDBSCAN(
    min_cluster_size=50,      # Minimum number of documents needed to form a cluster

    metric="euclidean",      # Use Euclidean distance to measure similarity

    cluster_selection_method="eom" # Use the EOM method to choose clusters
).fit(reduced_embeddings)

# Get the cluster label assigned to each document
clusters = hdbscan_model.labels_

print(clusters)

# How many clusters did we generate?
print(len(set(clusters)))
```

```
[137 137 -1 ... -1 128 -1]
154
```

```
# Convert the abstracts from the dataset format to a Python list
print(type(abstracts))
```

```
<class 'datasets.arrow_dataset.Column'>
```

```
#A Hugging Face Column does not accept numpy.int64 indices.
# It only accepts normal Python int.
# use int(index)
# ==== OR ====
# Convert the entire column to a list once
# abstracts = list(abstracts)
# Then use the original code
```

```
# Inspecting the Clusters
```

```
# Convert the abstracts into a Python list so they can be accessed by index.
abstracts = list(abstracts)
```

```
import numpy as np

# Select the cluster number to view
cluster = 0

# Find the first 3 abstracts that belong to the selected cluster
for index in np.where(clusters == cluster)[0][:3]:

    # Print the first 300 characters of each abstract
    print(abstracts[index][:300] + "... \n")
```

This works aims to design a statistical machine translation from English text to American Sign Language (ASL). The system is based on Moses tool with some modifications and the results are synthesized through a 3D avatar for interpretation. First, we translate the input text to gloss, a written fo...

Researches on signed languages still strongly dissociate linguistic issues related on phonological and phonetic aspects, and gesture studies for recognition and synthesis purposes. This paper focuses on the imbrication of motion and meaning for the analysis, synthesis and evaluation of sign lang...

Modern computational linguistic software cannot produce important aspects of sign language translation. Using some researches we deduce that the majority of automatic sign language translation systems ignore many aspects when they generate animation; therefore the interpretation lost the truth inf...

```
import pandas as pd
# Reduce 384-dimensional embeddings to two dimensions for easier visualization
reduced_embeddings = UMAP(
    n_components=2,      # Reduce to 2 dimension

    min_dist=0.0,      # Keep similar points close together

    metric="cosine",    # Use cosine distance
```

```

random_state=42      # Set a fixed random seed

).fit_transform(embeddings)

# Create a DataFrame with the reduced embeddings
df = pd.DataFrame(reduced_embeddings, columns=["x", "y"])

# Add the document titles
df["title"] = titles

# Add the cluster labels as strings
df["cluster"] = [str(c) for c in clusters]

# Select only the documents that belong to clusters
clusters_df = df.loc[df.cluster != "-1", :]

# Select only the outlier documents
outliers_df = df.loc[df.cluster == "-1", :]

print(clusters_df)
print(outliers_df)

```

```

/usr/local/lib/python3.12/dist-packages/umap/umap_.py:1952: UserWarning: n_jobs value 1 overridden to 1 by setting random_st
warn(

```

	x	y	title \
0	5.259873	-0.112428	Introduction to Arabic Speech Recognition Usin...
1	5.273234	-0.103197	Arabic Speech Recognition System using CMU-Sph...
4	4.323878	1.804335	Segmentation and Context of Literary and Music...
8	3.344908	1.641876	Learning Probabilistic Models of Word Sense Di...
11	-1.369643	0.841759	Bio-linguistic transition and Baldwin effect i...
...	...	...	...
44937	3.358378	1.649927	Selective Sampling for Example-based Word Sens...
44938	0.434781	-1.648514	Practical experiments with regular approximati...
44939	-3.817490	0.055345	Question Answering System Using Syntactic Info...
44941	-3.539318	-1.718429	Resolution of Indirect Anaphora in Japanese Se...
44947	3.962591	0.032059	HMM Specialization with Selective Lexicalization

	cluster
0	137
1	137
4	102
8	133
11	95
...	...
44937	133
44938	125
44939	89
44941	10
44947	128

[30178 rows x 4 columns]

	x	y	title \
2	4.755688	-0.237090	On the Development of Text Input Method - Less...
3	0.159493	-0.673342	Network statistics on early English Syntax: St...
5	3.506535	0.547703	International Standard for a Linguistic Annota...
6	3.417321	0.669793	A Formal Model of Dictionary Structure and Con...
7	-0.590658	-1.319472	Practical Approach to Knowledge-based Question...
...	...	...	...
44943	3.715235	-0.908435	An Estimate of Referent of Noun Phrases in Jap...
44944	3.718072	-0.950499	Resolution of Verb Ellipsis in Japanese Senten...
44945	3.623115	-1.104341	An Example-Based Approach to Japanese-to-Engli...
44946	-0.568136	-1.347879	Deduction over Mixed-Level Logic Representatio...
44948	-0.557082	-1.365685	Mixed-Level Knowledge Representation and Varia...

	cluster
2	-1
3	-1
5	-1
6	-1
7	-1
...	...
44943	-1
44944	-1
44945	-1
44946	-1
44948	-1

[14771 rows x 4 columns]

```

# Import the Matplotlib library for data visualization
import matplotlib.pyplot as plt

```

```

# Plot outliers and non-outliers separately
plt.scatter(outliers_df.x, outliers_df.y,

```

```

alpha=0.05,      # Make the points transparent

s=2,             # Set a small point size

c="grey")        # Use grey color for outliers

# Plot the clustered documents with different colors for each cluster
plt.scatter(
clusters_df.x, clusters_df.y,
c=clusters_df.cluster.astype(int),  # Color points based on cluster labels
alpha=0.6, s=2,
cmap="tab20b"      # Use the Tab20b color map
)

# Hide the x-axis and y-axis

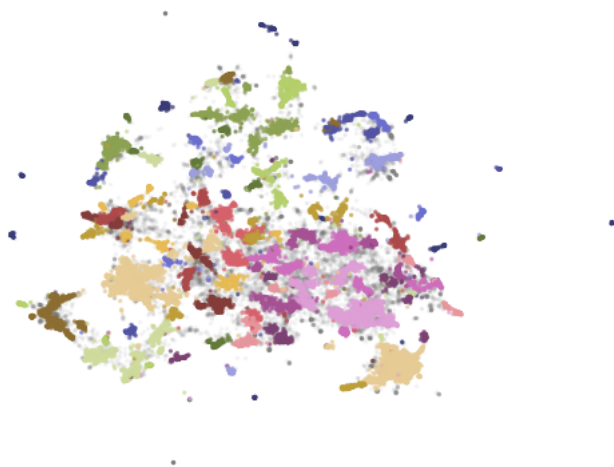
plt.axis("off")

```

```

(np.float64(-7.914401292800903),
 np.float64(11.045439004898071),
 np.float64(-7.8091048240661625),
 np.float64(11.462062740325928))

```



```
# From Text Clustering to Topic Modeling
```

```
# BERTopic
```

```
# Install BERTopic and the required libraries for topic modeling and text embeddings
!pip install bertopic umap-learn hdbscan sentence-transformers
```

```

Collecting bertopic
  Downloading bertopic-0.17.4-py3-none-any.whl.metadata (24 kB)
Requirement already satisfied: umap-learn in /usr/local/lib/python3.12/dist-packages (0.5.12)
Requirement already satisfied: hdbscan in /usr/local/lib/python3.12/dist-packages (0.8.44)
Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.12/dist-packages (5.6.0)
Requirement already satisfied: numpy>=1.20.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (2.0.2)
Requirement already satisfied: pandas>=1.1.5 in /usr/local/lib/python3.12/dist-packages (from bertopic) (2.2.2)
Requirement already satisfied: plotly>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (5.24.1)
Requirement already satisfied: scikit-learn>=1.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (1.6.1)
Requirement already satisfied: tqdm>=4.41.1 in /usr/local/lib/python3.12/dist-packages (from bertopic) (4.67.3)
Requirement already satisfied: llvmlite>0.36.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (0.43.0)
Requirement already satisfied: scipy>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from umap-learn) (1.16.3)
Requirement already satisfied: numba>=0.51.2 in /usr/local/lib/python3.12/dist-packages (from umap-learn) (0.60.0)
Requirement already satisfied: pynndescent>=0.5 in /usr/local/lib/python3.12/dist-packages (from umap-learn) (0.6.0)
Requirement already satisfied: joblib>=1.0 in /usr/local/lib/python3.12/dist-packages (from hdbscan) (1.5.3)
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in /usr/local/lib/python3.12/dist-packages (from sentence-trans
Requirement already satisfied: huggingface-hub>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transform
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.11
Requirement already satisfied: typing_extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transfor
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->sentence
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->
Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->s
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->sente
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->sentence
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.1.5->bert
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.1.5->bertopic) (202
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.1.5->bertopic) (2
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly>=4.7.0->bertopic) (
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.0->be
Requirement already satisfied: setuptools>=82 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-tran
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-tran
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-tr

```

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Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformer)
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Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolk
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-cusparselt-cu12==12.5.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-tool
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[c
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.4
Requirement already satisfied: tokenizers<0.21,>=0.20 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.4
Requirement already satisfied: cuda-pathfinder==1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1
```

```
# Import the BERTopic class
from bertopic import BERTopic

# Train our model with our previously defined models
topic_model = BERTopic(
    embedding_model=embedding_model, # Use the embedding model to generate document embeddings

    umap_model=umap_model,           # Use UMAP to reduce embedding dimensions

    hdbscan_model=hdbscan_model,      # Use HDBSCAN to group similar documents into topics

    verbose=True)                    # Show the training progress

# Input text documents
# Precomputed embeddings for documents
topic_model.fit(abstracts, embeddings)
```

```
2026-07-20 15:47:16,438 - BERTopic - Dimensionality - Fitting the dimensionality reduction algorithm
2026-07-20 15:48:45,994 - BERTopic - Dimensionality - Completed ✓
2026-07-20 15:48:45,998 - BERTopic - Cluster - Start clustering the reduced embeddings
2026-07-20 15:48:49,059 - BERTopic - Cluster - Completed ✓
2026-07-20 15:48:49,081 - BERTopic - Representation - Fine-tuning topics using representation models.
2026-07-20 15:48:53,419 - BERTopic - Representation - Completed ✓
```

```
# Display information about all the topics discovered by the BERTopic model
print(topic_model.get_topic_info())
```

	Topic	Count	Name \
0	-1	14771	-1_of_the_and_to
1	0	2113	0_speech_asr_recognition_end
2	1	2071	1_question_qa_answer_questions
3	2	1104	2_translation_nmt_machine_bleu
4	3	921	3_summarization_summaries_summary_abstractive
..	...	...	...
149	148	54	148_reviews_opinion_summaries_summarization
150	149	54	149_typing_entity_type_types
151	150	54	150_gans_gan_adversarial_generation
152	151	50	151_counterfactual_counterfactuals_cad_causal
153	152	50	152_factuality_factual_attribution_llms

	Representation \
0	[of, the, and, to, in, we, language, for, that...
1	[speech, asr, recognition, end, acoustic, spea...
2	[question, qa, answer, questions, answering, a...
3	[translation, nmt, machine, bleu, neural, engl...
4	[summarization, summaries, summary, abstractiv...
..	...
149	[reviews, opinion, summaries, summarization, r...
150	[typing, entity, type, types, grained, mention...
151	[gans, gan, adversarial, generation, generativ...
152	[counterfactual, counterfactuals, cad, causal,...
153	[factuality, factual, attribution, llms, facts...

	Representative Docs
0	[Natural language tasks like Named Entity Re...
1	[Recent advances in text-to-speech (TTS) led...
2	[With the development of deep learning techn...
3	[A prerequisite for training corpus-based ma...
4	[Pre-trained neural abstractive summarizatio...
..	...
149	[When faced with a large number of product r...

```

150 [ For the task of fine-grained entity typing ...
151 [ Text generation is of particular interest i...
152 [ For text classification tasks, finetuned la...
153 [ Assessing factuality of text generated by l...

```

[154 rows x 5 columns]

```

# Display the keywords and their scores for Topic 0
print(topic_model.get_topic(0))

```

```

# Find topics that are most similar to the given search text
print(topic_model.find_topics("topic modeling"))

```

```

# Display the keywords and their scores for Topic 22
print(topic_model.get_topic(22))

```

```

# Display the topic assigned to the specified document title
print(topic_model.topics_[titles.index("BERTopic: Neural topic modeling with a class-based TF-IDF procedure")])

```

```

[('speech', np.float64(0.029006046231374278)), ('asr', np.float64(0.0192572104101309)), ('recognition', np.float64(0.0136982
([19, -1, 53, 31, 83], [np.float32(0.9536687), np.float32(0.91216874), np.float32(0.9072217), np.float32(0.90511006), np.flo
[('attention', np.float64(0.03453290175573789)), ('transformer', np.float64(0.027178794837357474)), ('transformers', np.floa
19

```

```

# Convert the titles into a Python list
titles=list(titles)

```

```

# Visualize the documents and their topics in a 2D plot
fig = topic_model.visualize_documents(
    titles,                                # Document titles to display

```

```

    reduced_embeddings=reduced_embeddings, # 2D embeddings for visualization

```

```

    width=1200,                            # Set the width of the plot

```

```

    hide_annotations=True)                 # Hide document labels for a cleaner view

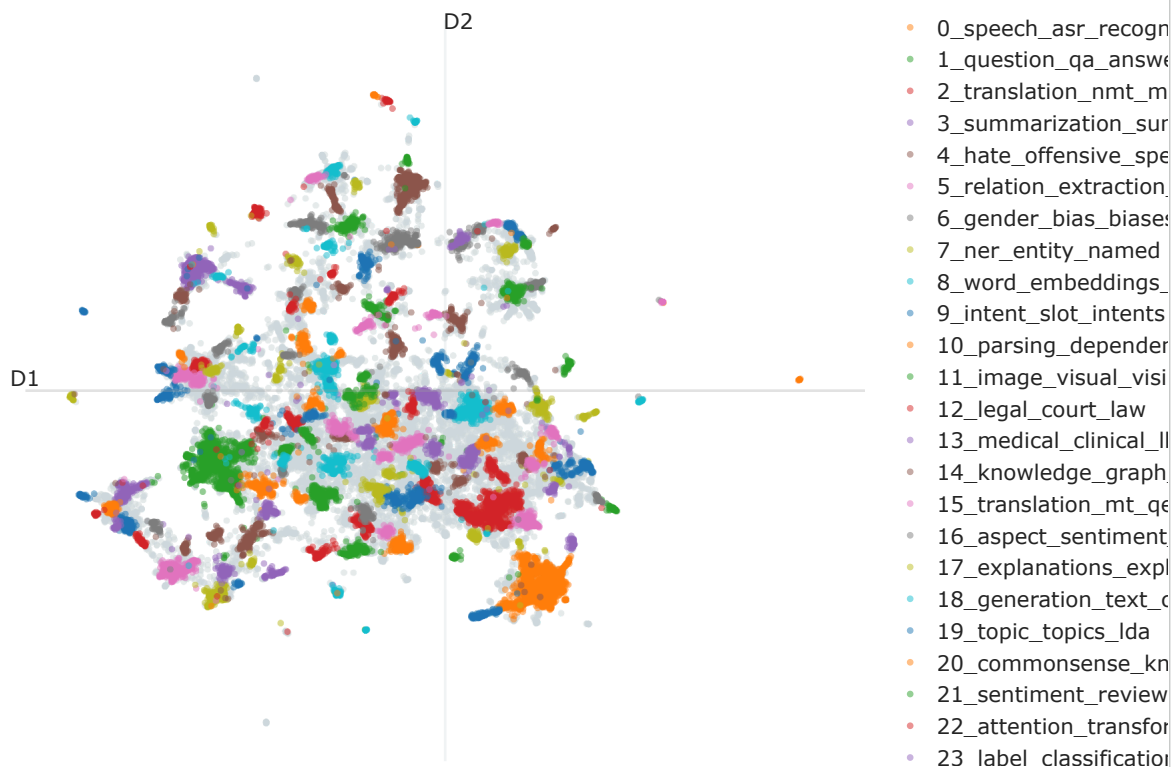
```

```

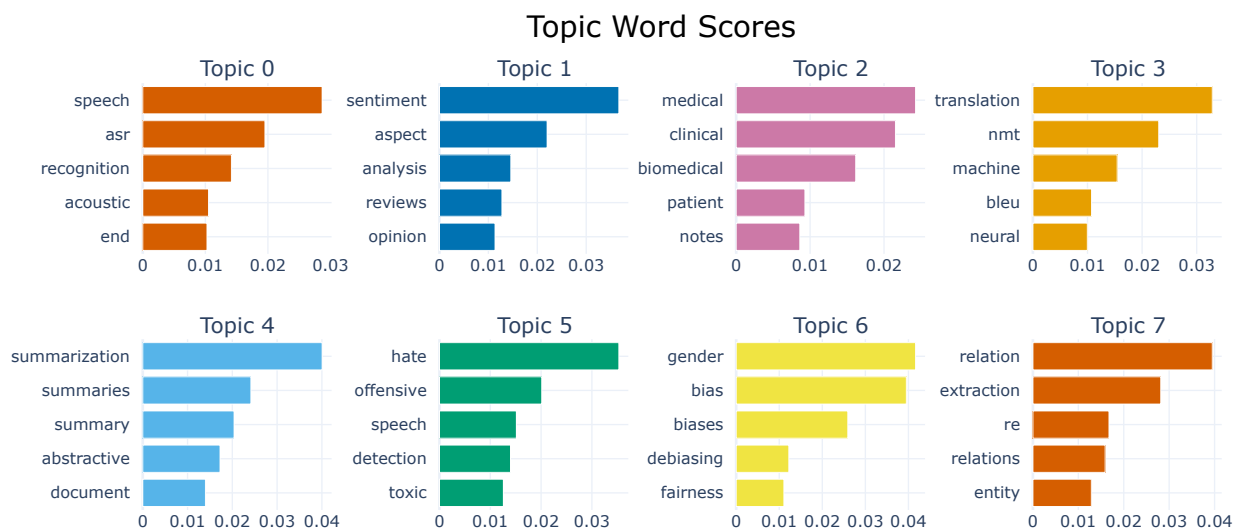
# Update fonts of legend for easier visualization
fig.update_layout(font=dict(size=16))

```

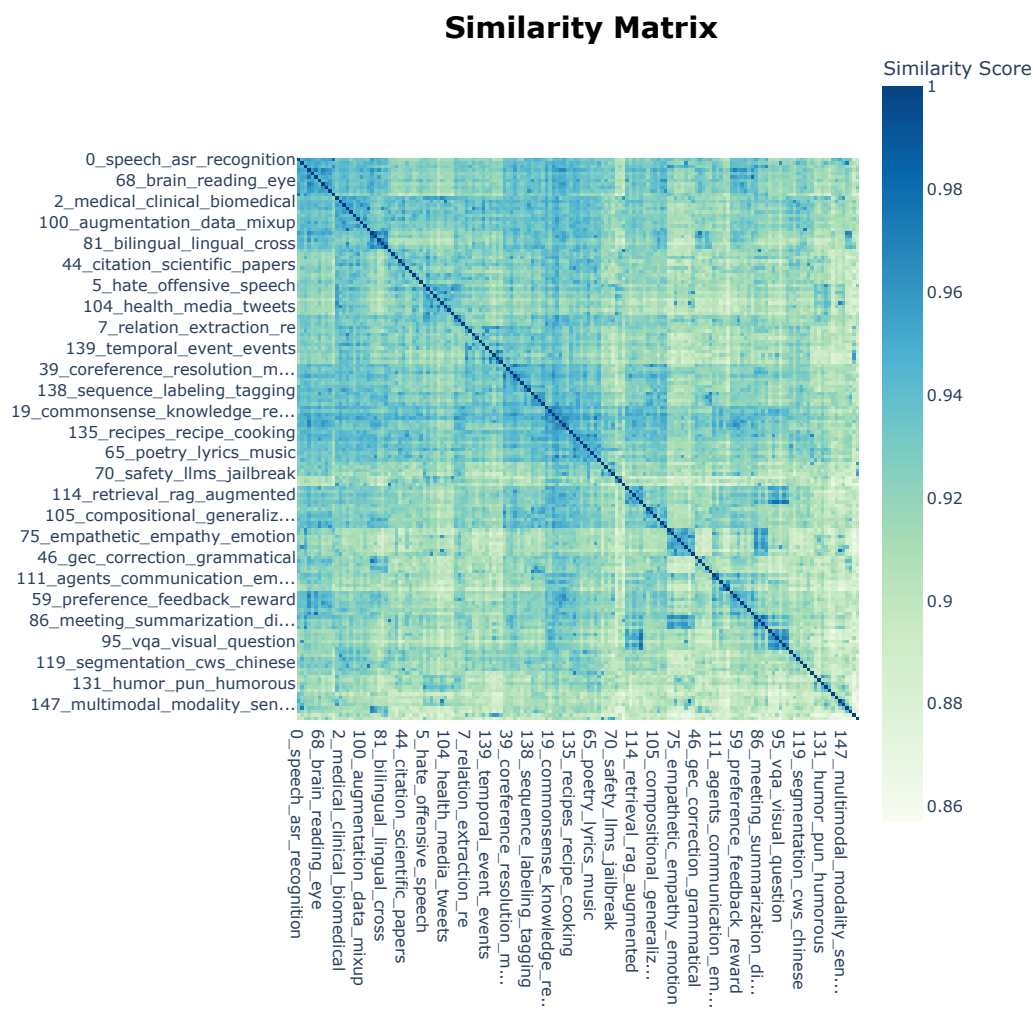
Documents and Topics




```
# Visualize barchart with ranked keywords
topic_model.visualize_barchart()
```

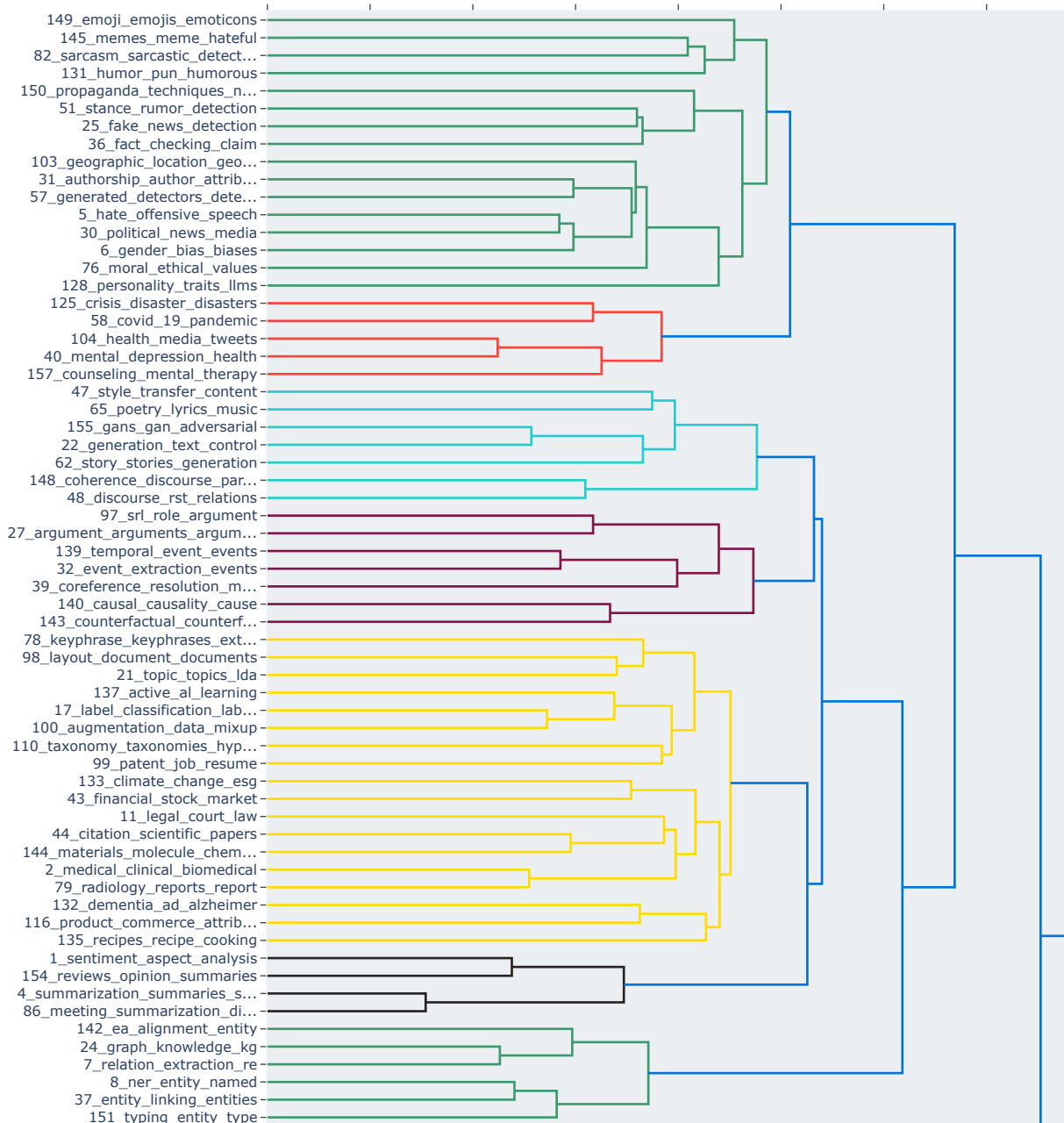


```
# Visualize relationships between topics
topic_model.visualize_heatmap(n_clusters=30)
```



```
# Visualize the potential hierarchical structure of topics
topic_model.visualize_hierarchy()
```


Hierarchical Clustering



Adding a Special Lego Block

```
import pandas as pd
from copy import deepcopy

# Save original topic representations
original_topics = deepcopy(topic_model.topic_representations_)

def topic_differences(model, original_topics, nr_topics=5):
    """Show the differences in topic representations between two models."""

    df = pd.DataFrame(columns=["Topic", "Original", "Updated"])

    for topic in range(nr_topics):
        # Extract top 5 words per topic from the original model
        og_words = " | ".join(list(zip(*original_topics[topic])))[0][:5])

        # Extract top 5 words per topic from the updated model
        new_words = " | ".join(list(zip(*model.get_topic(topic))))[0][:5]

        # Add a new row to the DataFrame
        df.loc[len(df)] = [topic, og_words, new_words]

    return df
topic_differences(topic_model, original_topics)
```

Topic	Original	Updated
0	speech asr recognition end acoustic	speech asr recognition end acoustic
1	question qa answer questions answering	question qa answer questions answering
2	translation nmt machine bleu neural	translation nmt machine bleu neural
3	summarization summaries summary abstract...	summarization summaries summary abstract...
4	hate offensive speech detection toxic	hate offensive speech detection toxic

KeyBERTInspired

```
# Import the KeyBERTInspired representation model
from bertopic.representation import KeyBERTInspired

# Create a KeyBERTInspired model to generate better topic representations
representation_model = KeyBERTInspired()

# Update the topic representations using the KeyBERTInspired model
topic_model.update_topics(abstracts, # inputs
                           representation_model=representation_model)

# Compare the updated topics with the original topics
topic_differences(topic_model, original_topics)
```

Topic	Original	Updated
0	speech asr recognition end acoustic	language speech phonetic translation t...
1	question qa answer questions answering	answering questions comprehension answer...
2	translation nmt machine bleu neural	translation translate translations trans...
3	summarization summaries summary abstract...	summarization summarizers summaries summ...
4	hate offensive speech detection toxic	hate hateful cyberbullying language of...

Maximal marginal relevance

```
# Import the Maximal Marginal Relevance representation model
from bertopic.representation import MaximalMarginalRelevance

# Create a Maximal Marginal Relevance model to generate more diverse topic keywords
representation_model = MaximalMarginalRelevance(diversity=0.2) # Set the diversity level for selecting keywords

# Update the topic representations using the MMR model
topic_model.update_topics(abstracts,
                           representation_model=representation_model)

# Show topic differences
topic_differences(topic_model, original_topics)
```

Topic	Original	Updated
0	speech asr recognition end acoustic	speech asr audio model automatic
1	question qa answer questions answering	retrieval comprehension passage hop kn...
2	translation nmt machine bleu neural	translation nmt bleu neural languages
3	summarization summaries summary abstract...	summarization summaries extractive evalu...
4	hate offensive speech detection toxic	hate offensive toxic abusive hateful

The Text Generation Lego Block

```
# Import the Hugging Face pipeline and BERTopic TextGeneration representation model
from transformers import pipeline
from bertopic.representation import TextGeneration

prompt = ""
I have a topic that contains the following documents:
[DOCUMENTS]

The topic is described by the following keywords:
'[KEYWORDS]'.
```

Based on the documents and keywords, what is this topic about?
"""

```
# Load Flan-T5
generator = pipeline(
    "text2text-generation",
    model="google/flan-t5-small"
)

# Create representation model
representation_model = TextGeneration(
    generator,
    prompt=prompt,
    doc_length=50,
    tokenizer="whitespace" # Split text using whitespace
)

# Update topic representations
topic_model.update_topics(
    abstracts,
    representation_model=representation_model
)

# Compare topics
topic_differences(topic_model, original_topics)
```

config.json: 1.40k/? [00:00<00:00, 135kB/s]

model.safetensors: 100% 308M/308M [00:02<00:00, 169MB/s]

generation_config.json: 100% 147/147 [00:00<00:00, 8.22kB/s]

tokenizer_config.json: 2.54k/? [00:00<00:00, 172kB/s]

spiece.model: 100% 792k/792k [00:00<00:00, 3.81MB/s]

tokenizer.json: 2.42M/? [00:00<00:00, 52.0MB/s]

special_tokens_map.json: 2.20k/? [00:00<00:00, 181kB/s]

Hardware accelerator e.g. GPU is available in the environment, but no `device` argument is passed to the `Pipeline` object.
100%[██████████] 154/154 [01:08<00:00, 2.26it/s]

Topic		Original	Updated
0	0	speech asr recognition end acoustic	Automated Speech Recognition
1	1	question qa answer questions answering	Multi-hop question answering
2	2	translation nmt machine bleu neural	Science/Tech
3	3	summarization summaries summary abstract...	Summarization
4	4	hate offensive speech detection toxic	Science/Tech

Start coding or [generate](#) with AI.

```
# Install or update BERTopic and its required libraries
!pip install -U bertopic sentence-transformers umap-learn hdbscan
```

Requirement already satisfied: numpy>=1.20.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (2.0.2)
Requirement already satisfied: pandas>=1.1.5 in /usr/local/lib/python3.12/dist-packages (from bertopic) (2.2.2)
Requirement already satisfied: plotly>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (5.24.1)
Requirement already satisfied: scikit-learn>=1.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (1.6.1)
Requirement already satisfied: tqdm>=4.41.1 in /usr/local/lib/python3.12/dist-packages (from bertopic) (4.67.3)
Requirement already satisfied: llvmlite>0.36.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (0.43.0)
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (4.41.0)
Requirement already satisfied: huggingface-hub>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (0.23.0)
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.11.0)
Requirement already satisfied: scipy>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.16.0)
Requirement already satisfied: typing_extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (4.5.0)
Requirement already satisfied: numba>=0.51.2 in /usr/local/lib/python3.12/dist-packages (from umap-learn) (0.60.0)
Requirement already satisfied: pynndescent>=0.5 in /usr/local/lib/python3.12/dist-packages (from umap-learn) (0.6.0)
Requirement already satisfied: joblib>=1.0 in /usr/local/lib/python3.12/dist-packages (from hdbscan) (1.5.3)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->sentence-transformers) (3.12.2)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->sentence-transformers) (2023.5.0)
Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->sentence-transformers) (23.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->sentence-transformers) (6.0.1)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->sentence-transformers) (2.31.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.1.5->bertopic) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.1.5->bertopic) (2020.1)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.1.5->bertopic) (2022.7)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly>=4.7.0->bertopic) (8.1.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.0->bertopic) (3.1.0)

```
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,c
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->s
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0
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Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->se
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-tran
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolki
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-t
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolki
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolki
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolk
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-cusparselt-cu12==12.5.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-tool
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-too
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[c
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.4
Requirement already satisfied: tokenizers<0.21,>=0.20 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.4
Requirement already satisfied: cuda-pathfinder~1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.11.0->sen
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->hugging
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->huggingface-hub>=0.
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->huggingface-h
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->huggingface-h
```

```
# Unable to do in openai . it needs billing
```

```
# Import the OpenAI library and BERTopic OpenAI representation model
import openai
from bertopic.representation import OpenAI

# Define the prompt for generating short topic labels
prompt = """
I have a topic that contains the following documents:
[DOCUMENTS]
The topic is described by the following keywords: [KEYWORDS]
Based on the information above, extract a short topic label in
the following format:
topic: <short topic label>
"""

# Create an OpenAI representation model for generating topic labels
client = openai.OpenAI(api_key="api_key")
representation_model = OpenAI(
    client, model="gpt-3.5-turbo", # OpenAI model used for topic labeling

    exponential_backoff=True,      # Retry automatically if a request fail

    chat=True,                    # Use the chat completion API

    prompt=prompt                 # Prompt used to generate topic labels
)

# Update the topic representations using the OpenAI model
topic_model.update_topics(abstracts,
representation_model=representation_model)

# Show topic differences
topic_differences(topic_model, original_topics)
```

```
# Visualize the documents and topics using DataMapPlot
fig = topic_model.visualize_document_datamap(
    titles,                        # Document titles to display
    topics=list(range(20)),        # Show the first 20 topics
    reduced_embeddings=reduced_embeddings, # 2D embeddings for visualization
    width=1200,                   # Set the width of the plot
    label_font_size=11,           # Set the font size of topic labels
    label_wrap_width=20,          # Wrap long labels after 20 characters
    use_medoids=True              # Use the central document to represent each topic
)
```